COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 007	**SECTION:**	CPE11S4
DATE PERFORMED:	8/05/2026	**DATE SUBMITTED:**	8/05/2026
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ACTIVITY NO. 2.1 Control Structures (part 1)

1. PROCEDURE OUTPUT

1. Develop a C++ program that will determine determine if a department store customer has exceeded the credit limit on a charge account. For each customer, the following facts are available:
a. Account number
b. Balance at the beginning of the month
c. Total of all items charged by this customer this month
d. Total of all credits applied to this customer's account this month
e. Allowed credit limit
The program should input each of these facts, calculate the new balance (=beginning balance +charges -credits), and determine if the new balance exceeds the customer's credit limit. For those customers whose credit limit is exceeded, the program should display the customer's account number, credit limit, new balance and the message "Credit limit exceeded."

main.cpp

```cpp
#include <iostream>
using namespace std;

int main () {

    double acc, bal, nbal, chrg, cred, lim;
    cout << "Please fill out the following (-1 on Account Number to stop)." << endl;
    cout << "Account Number: ";
    cin >> acc;


    while ( acc != -1) {
    cout << "Beginning balance: PHP ";
    cin >> bal;
    cout << "Total charge of all items: PHP ";
    cin >> chrg;
    cout << "Credits applied: PHP ";
    cin >> cred;
    cout << "Credit card limit: PHP ";
    cin >> lim;

    nbal = (bal + chrg) - cred;

    if (nbal < lim) {
        cout << "Account Number: " << acc << endl;
        cout << "Did not exceed credit card limit." << endl;

    } else if (nbal > lim) {
        cout << "Account Number: " << acc << endl;
        cout << "Credit Limit: PHP " << lim << endl;
        cout << "New Balance: PHP " << nbal << endl;
        cout << "Credit limit exceeded.";

    }

    cout << endl;
    cout << "Account Number: ";
    cin >> acc;

    }
    cout << "Thank you for using this program, no more accounts to process." << endl;
    return 0;
}
```

Output

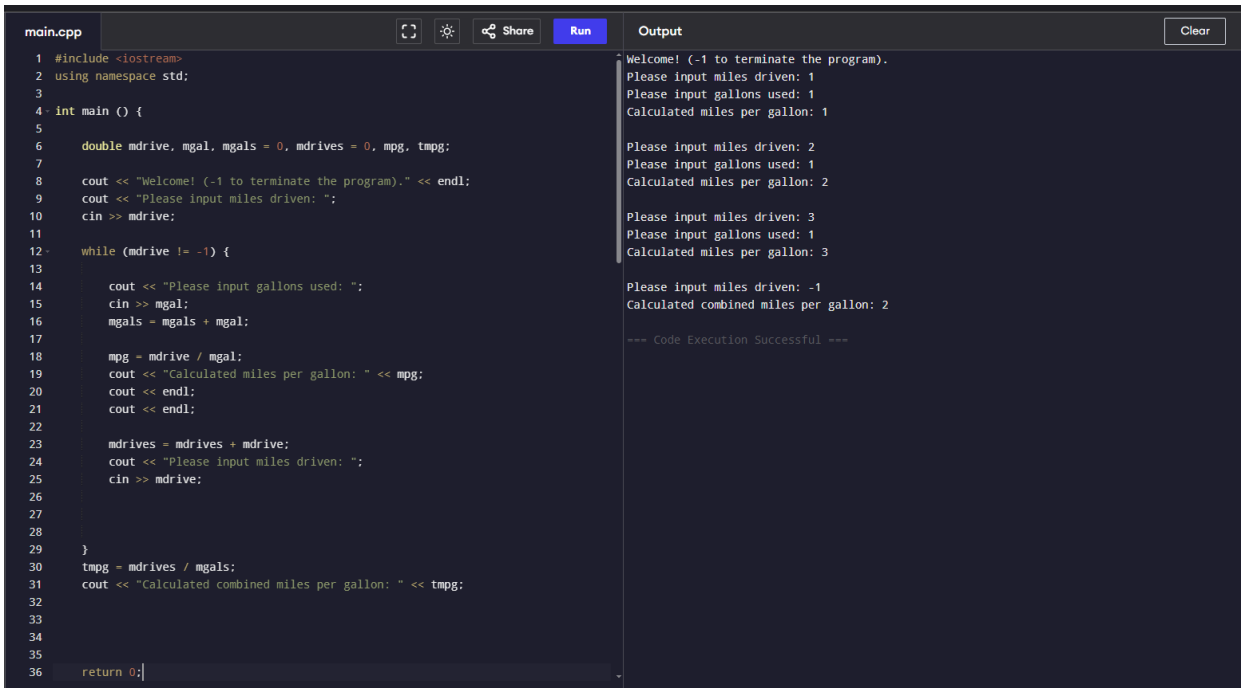
```
Please fill out the following (-1 on Account Number to stop).
Account Number: 123456
Beginning balance: PHP 100000
Total charge of all items: PHP 12321
Credits applied: PHP 2431
Credit card limit: PHP 500000
Account Number: 123456
Did not exceed credit card limit.

Account Number: 41422322
Beginning balance: PHP 12000
Total charge of all items: PHP 5999
Credits applied: PHP 7999
Credit card limit: PHP 2000
Account Number: 4.14223e+07
Credit Limit: PHP 2000
New Balance: PHP 10000
Credit limit exceeded.
Account Number:
```

This program checks if an account exceeds its credit card limit. It calculates the new balance using the beginning balance, total charges, and credits applied, then displays the result. It continues until the user enters -1 as the account number.

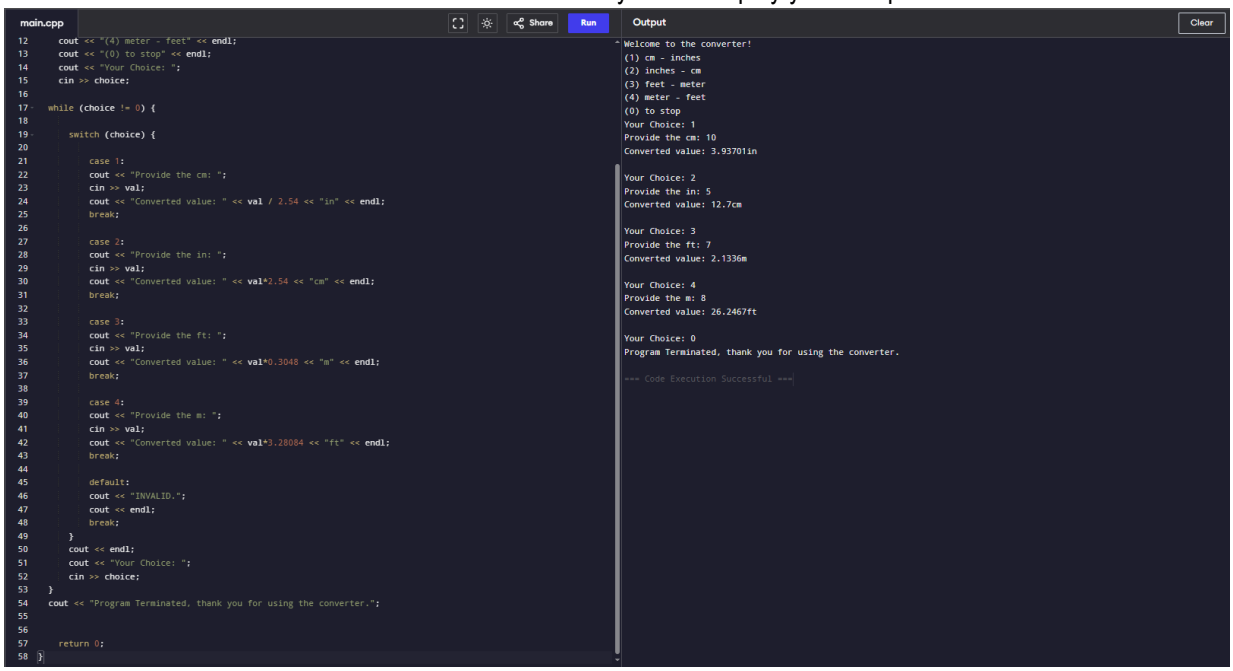
2. ANSWERS TO SUPPLEMENTAL ACTIVITY

1. Because of the price of gasoline, drivers are concerned with the mileage obtained by their automobiles. One driver has kept track of several tankfuls of gasoline by recording miles driven and gallons used for each tankful. Develop a program that will input the miles driven and gallons used for each tankful. The program should calculate and display the miles per gallon obtained for each tankful. After processing all input information, the program should calculate and print the combined miles per gallon obtained for all tankful.



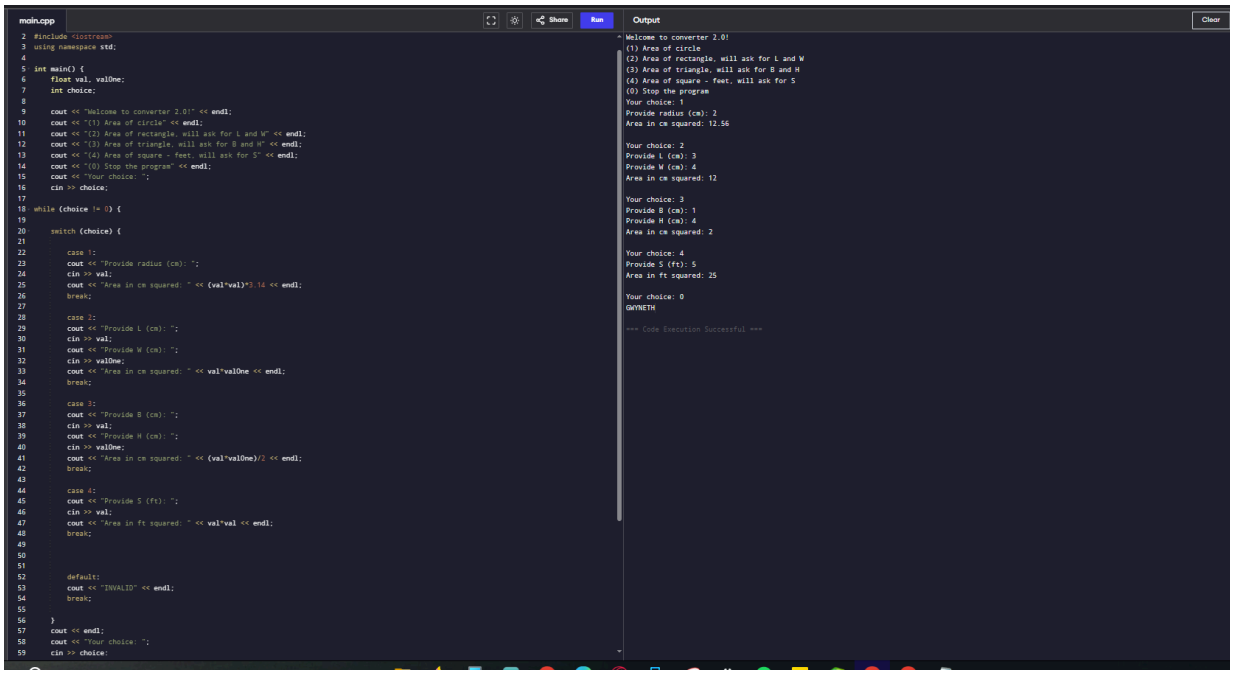
I made this program to calculate the miles per gallon for each tankful of gasoline. It keeps asking for the miles driven and gallons used until I enter -1 to stop the program. It also adds all the miles and gallons together to get the combined miles per gallon. Through this activity, I learned how to use a while loop and running totals in C++.

3. Write a program that displays a menu for simple conversions such as the following:
(1) cm - inches
(2) inches - cm
(3) feet - meter
(4) meter - feet
Once selected, the user will be asked to enter a float to be converted. After the conversion, the user would be asked to convert another until the user doesn't want any more. Display your samples for all conversions



This program converts measurements between centimeters, inches, feet, and meters. It lets the user choose a conversion, enter a value, and repeats until the user chooses to stop.

4. Write a program that displays a menu for simple computation of formula such as the following:
(1) Area of circle, will ask for radius from user
(2) Area of rectangle, will ask for L and W
(3) Area of triangle, will ask for B and H
(4) Area of square - feet, will ask for S
Once selected, the user will be asked to enter a float and be computed as selected. After the computation, the user would be asked to compute another until the user doesn't want anymore. Display your samples for all conversions



This program calculates the area of different shapes, including a circle, rectangle, triangle, and square. It allows the user to choose a shape, enter the required measurements, and repeats until the user decides to stop.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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